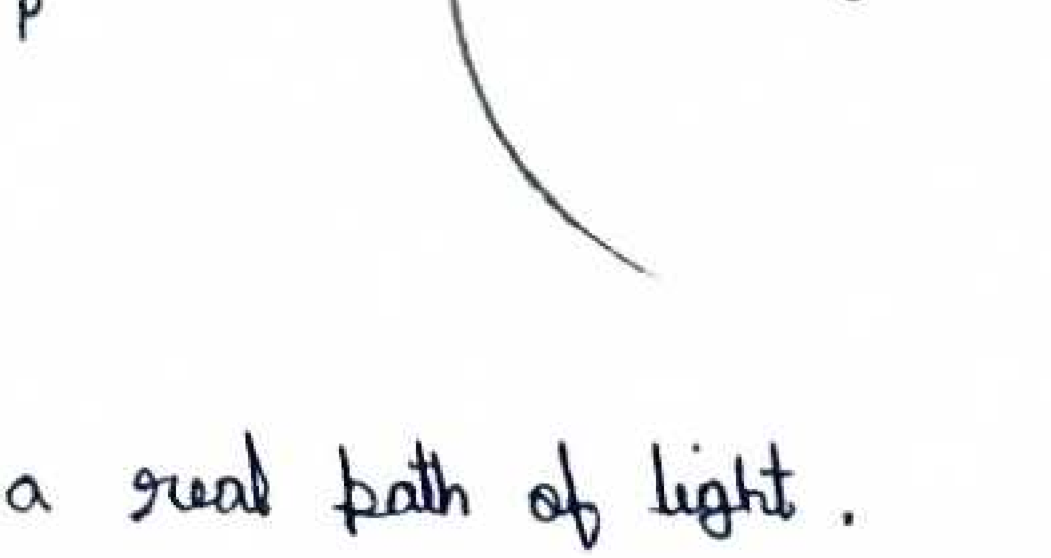
Q_1 State and Explain Fermat's principle of extremum path and use it to establish the laws of reflection and refraction.

Ans Fermat's Principle states that -

When a ray of light goes from one point to another through a set of media, it always follows a path along which the time taken is minimum.

Consider the fig below, where P and Q are 2 different points in different medium.



Let PAQ be a real path of light.

Let the speed of light in medium 1 is V_1

Let the speed of light in medium 2 is V_2

Then,

The time taken by light in reaching from P to Q along this path is given by -

$$t = \frac{PA}{V_1} + \frac{AQ}{V_2} \quad \text{—} \quad (1)$$

Now, a neighbouring path is PBQ which may be expressed in terms of parametre S, which could be either the lateral distance AB or angle APB or anything which shows a lateral shift of path.

Now, Mathematically

Fermat's principle states that $\frac{dt}{ds} = 0$

In 3d space, $\left(\frac{dt}{ds_1}\right)_{s_2} = 0$ and $\left(\frac{dt}{ds_2}\right)_{s_1} = 0$

Again,

$$eq\ (1) \Rightarrow t = \frac{PA}{V_1} + \frac{AQ}{V_2}$$

$$t = \frac{1}{c}\left[\frac{c \times PA}{V_1} + c \times \frac{AQ}{V_2}\right]$$

$$\boxed{\because \ \frac{c}{v} = \mu}$$

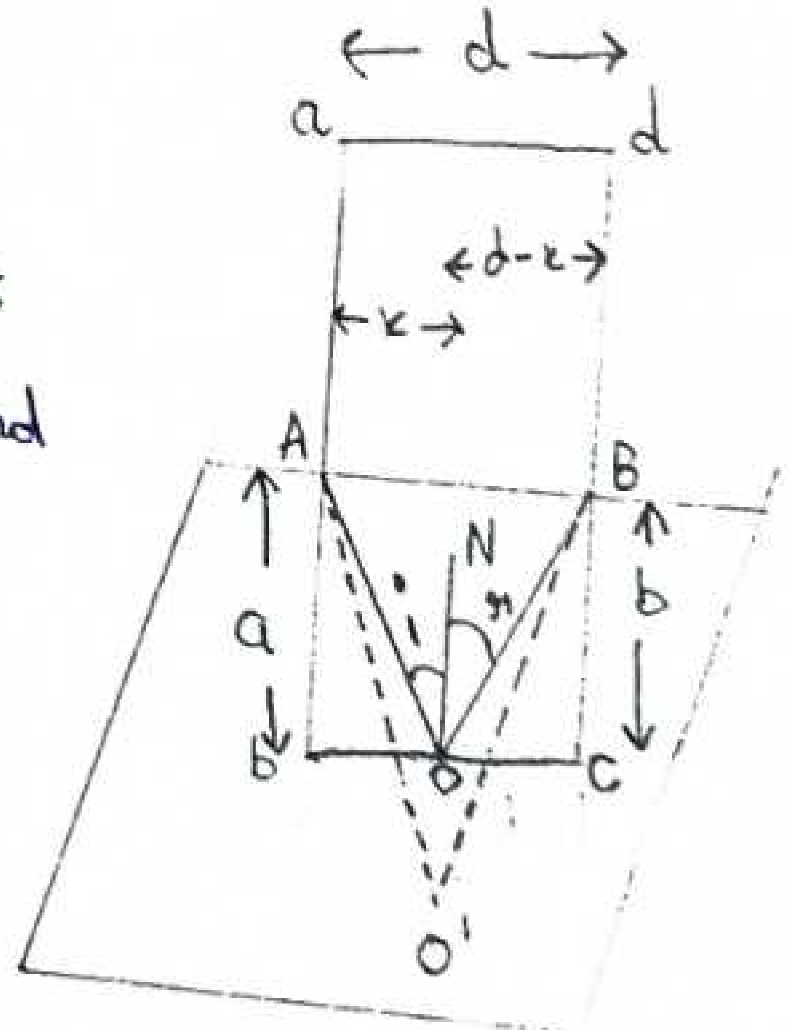
$$t = \frac{1}{c}\left[\mu_1 PA + \mu_2 AQ\right]$$

The quantity $(\mu_1 PA + \mu_2 AQ)$ is often called as the optical path. If we represent it by p,

Then, $$\frac{dp}{ds} = 0$$

This states fermat's principle in terms of optical path instead of time t.

Laws of Reflection using fermat's principle



Consider a plane mirror M_1M_2 and a plane abcd normal to it. A ray of light AO incident along OB. The rays AO, OB and normal ON lie in the same plane abcd. Consider another ray A'O reflected from point O' where OO' is normal to abcd.

$\therefore$ $AO' > AO$ and $O'B > OB$.

Thus, $(AO' + O'B) > (AO + OB)$, which is against Fermat's principle

Hence O' must coincide with O for the path to be minimum. This proves the 1^{st} law of reflection

Now, For proving 2nd law,

$$t = \frac{AO + OB}{c} = \frac{\sqrt{a^2 + x^2} + \sqrt{b^2 + (d-x)^2}}{c}$$

If the point O is displaced slightly along M_1 and M_2, x will change by dx and slightly different path will be obtained for light and time will be different. As A and B are fixed d will remain the same for both path.

So as per Fermat's principle $\frac{dt}{dx} = 0$

$$\frac{dt}{dx} = \frac{1}{c}\left[\frac{x}{\sqrt{a^2+x^2}} - \frac{d-x}{\sqrt{b^2+(d-x)^2}}\right] = 0$$

$$\frac{x}{\sqrt{a^2+x^2}} = \frac{d-x}{\sqrt{b^2+(d-x)^2}}$$

$$\sin i = \sin r$$

$$i = r$$

This proves the second law of reflection.

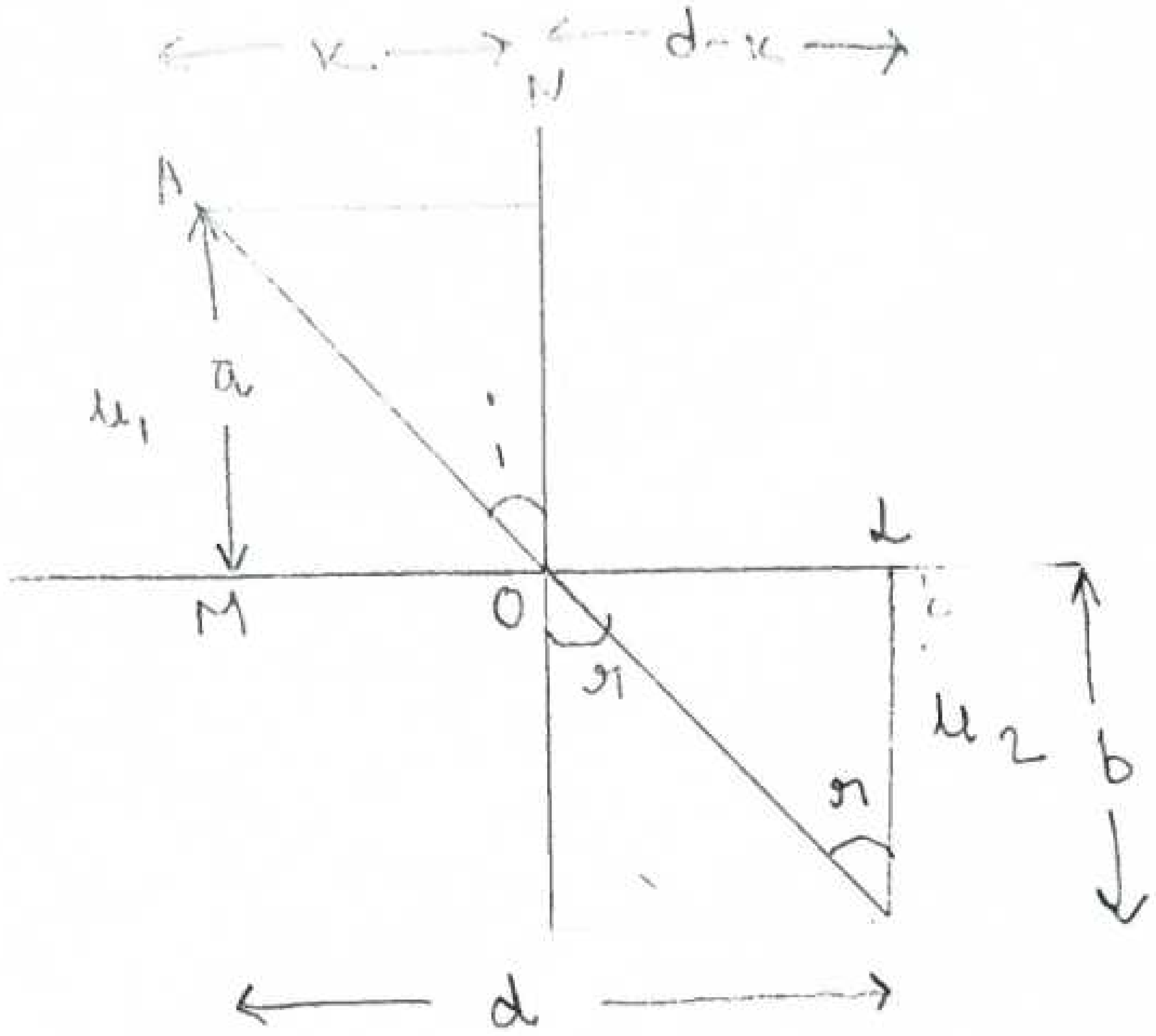
Laws of Refraction from fermat's principle

AOB is the path of a ray from medium of refractive index μ_1 into a medium of refractive index μ_2. As shown in the diagram passing via O at the interface where $\mu_2 > \mu_1$.

If c_1 is the velocity of light in the medium of refractive index μ_1 and c_2 that in medium of refractive index μ_2.

Then
$$t = \frac{AO}{c_1} + \frac{OB}{c_2}$$

$$t = \frac{\sqrt{a^2+x^2}}{c_1} + \frac{\sqrt{b^2+(d-x)^2}}{c_2}$$



For t to be minimum, $\dfrac{dt}{dx} = 0$

$$\Rightarrow \quad \frac{1}{c_1}\frac{x}{\sqrt{a^2+x^2}} + \frac{1}{c_2}\frac{d-x}{\sqrt{b^2+(d-x)^2}}$$

$$\frac{\sin i}{c_1} = \frac{\sin r}{c_1}$$

Now $\dfrac{c}{c_1} = \mu_1$ and $\dfrac{c}{c_2} = \mu_2$

So,
$$\frac{\mu_1}{c}\sin i = \frac{\mu_2}{c}\sin r$$

$$\Rightarrow \quad \mu_1 \sin i = \mu_2 \sin r$$

This proves the Snell's law of refraction - The 2nd law.
The 1st law could be proved same as to 1st law of reflection.

Q2 Derive Newton's lens formula. Also Explain Lagrange's eq of Magnification.

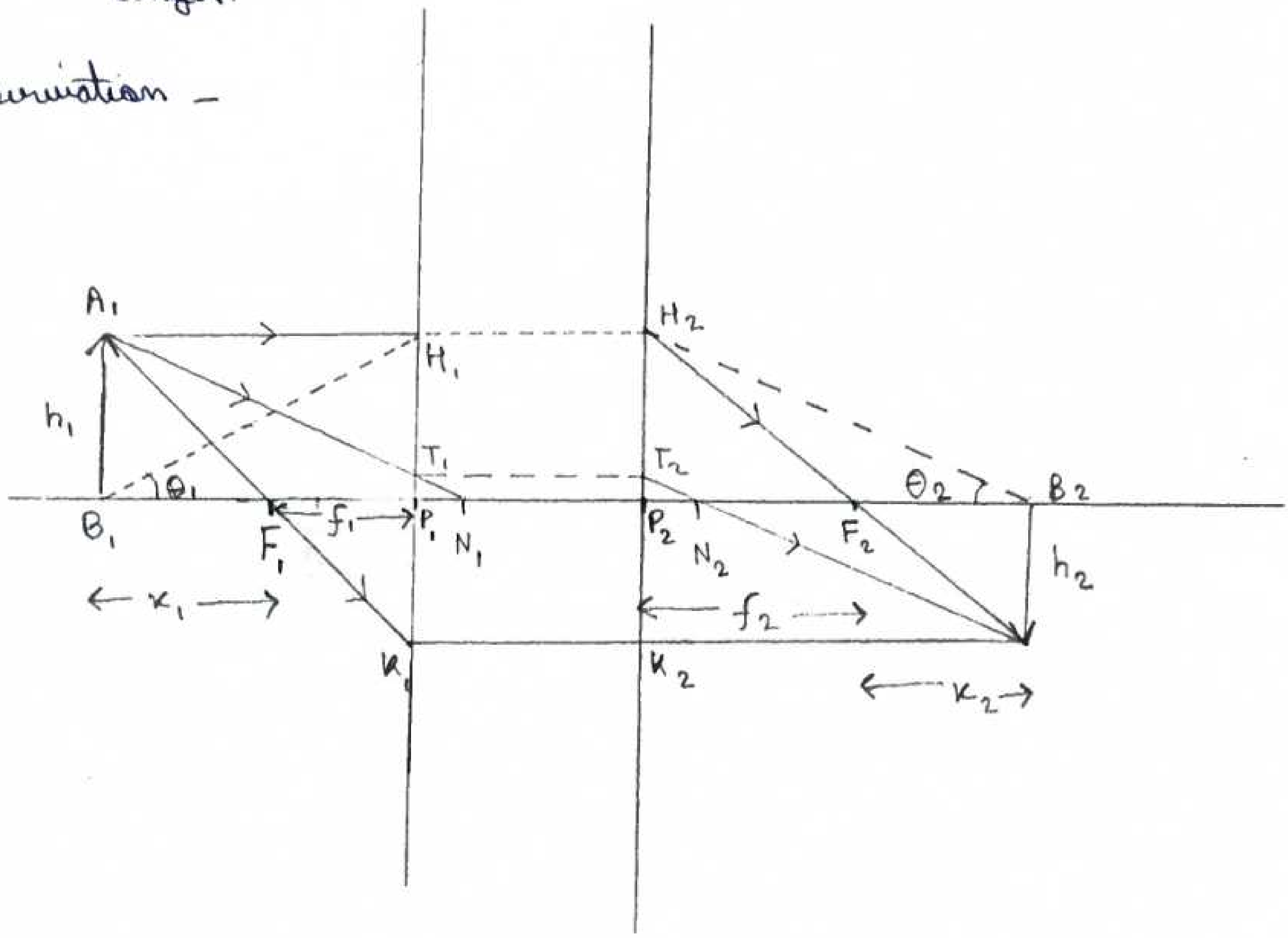
Ans Newton's formula states that $x_1 x_2 = f_1 f_2$

where x_1 = distance of object from first principle focus

x_2 = distance of object from second principle focus.

and f_1 and f_2 are the first and second principal focal length.

Derivation –



Consider this diagram where –

P_1 and P_2 = Principle points

N_1 and N_2 = Nodal Points

F_1 and F_2 = Focal points

To Find the image of object A_1B_1 draw a ray A_1H_1 parallel to principal axis touching the first principal plane at H_1. The Conjugated ray will proceed from H_2 a point on the second principal plane such that $H_2P_2 = H_1P_1$. It will pass through the second principle focus F_2.

a secondary $A_1T_1N_1$ directed towards the first Nodal point N_1 after refraction will pass through the second nodal point N_2 such that the ray $T_2N_2A_2$ is parallel to $A_1T_1N_1$.

Another ray A_1FK_1 passing through the first principle focus F_1 touches the first principle point plane at K_1 such that $K_2P_2 = K_1P_1$ and proceeds parallel to the principle axis.

Now, $\Delta S\ A_1B_1F_1$ and $P_1K_1F_1$ are similar

$$\frac{P_1K_1}{A_1B_1} = \frac{F_1P_1}{B_1F_1}$$

$P_1K_1 = P_2K_2 = A_2B_2$; $F_1P_1 = f_1$ and $B_1F_1 = x_1$

$$\frac{A_2B_2}{A_1B_1} = \frac{f_1}{x_1}$$

Also, $\Delta\ A_2B_2F_2$ and $\Delta H_2P_2F_2$ are similar

$$\frac{A_2B_2}{H_2P_2} = \frac{B_2F_2}{P_2F_2}$$

$H_2P_2 = H_1P_1 = A_1B_1$ and $F_2P_2 = f_2$ and $B_2F_2 = x_2$

$$\frac{A_2B_2}{A_1B_1} = \frac{x_2}{f_2} = \frac{h_2}{h_1} = m$$

and Hence, $x_1 x_2 = f_1 f_2$

This proves / Derived the newton's formula

Lagrange's law -

Mathematically, Lagrange's law stated that -

$$\mu_1 y_1 \tan\theta_1 = \mu_2 y_2 \tan\theta_2$$

Consider the ray diagram below which shows an object of y_1 is placed before another lens material of refractive index μ_2. The spherical surface's pole is at P and the image formed inside is of height y_2.

So,

μ_1 = refractive index of air

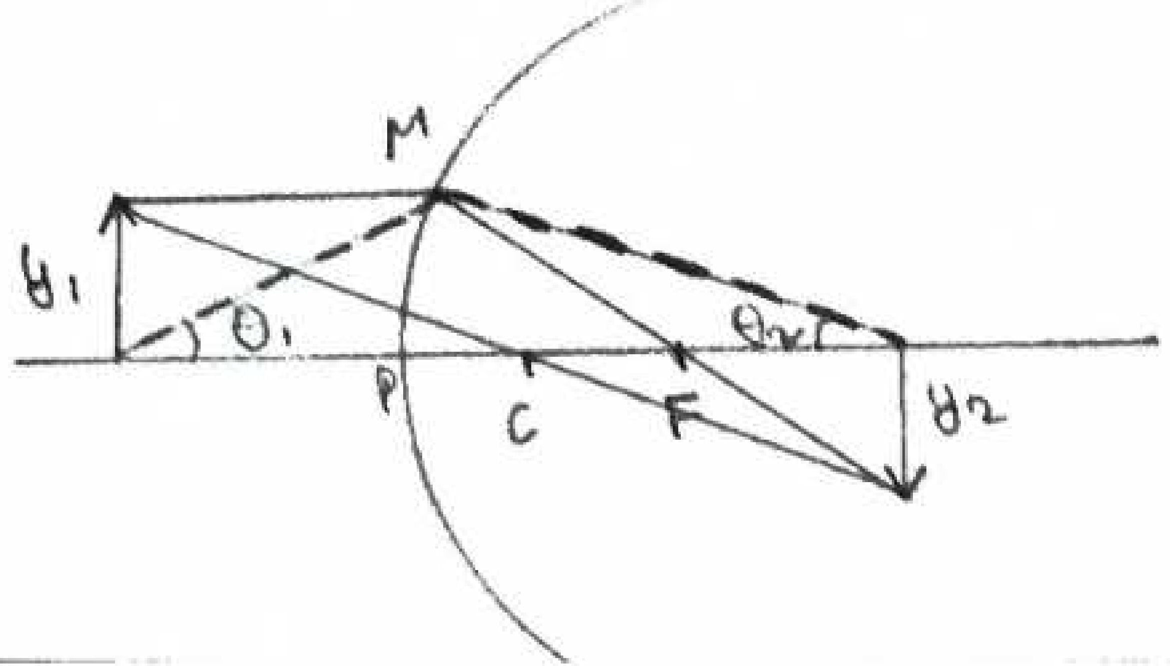
y_1 = height of object

$\tan\theta_1$ = angle btw principle axis and spherical surface point where incident ray hit.

μ_2 = refractive index of glass

y_2 = height of image

$\tan\theta_2$ = angle btw principle axis and the ray making the image at B,



Q3 Define Cardinal Points of a Co-axial Optical System. Also construct the image Using Cardinal points.

Ans In a system of lenses or mirrors or both, which forms ideal images of objects, there are 6 Cardinal points. These are -

1) Two Focal points
2) Two Principal points
3) Two Nodal points

a) Two Focal Points - A ray of light incident on the lens in the direction parallel to the principal axis on the side object side is bent on emergence in such a way that it passes through F_2 the point on the axis on the image side. Similarly a ray of light passing through the point F_1 on the axis, on the object side is rendered parallel to the axis after emergence Through the lens.

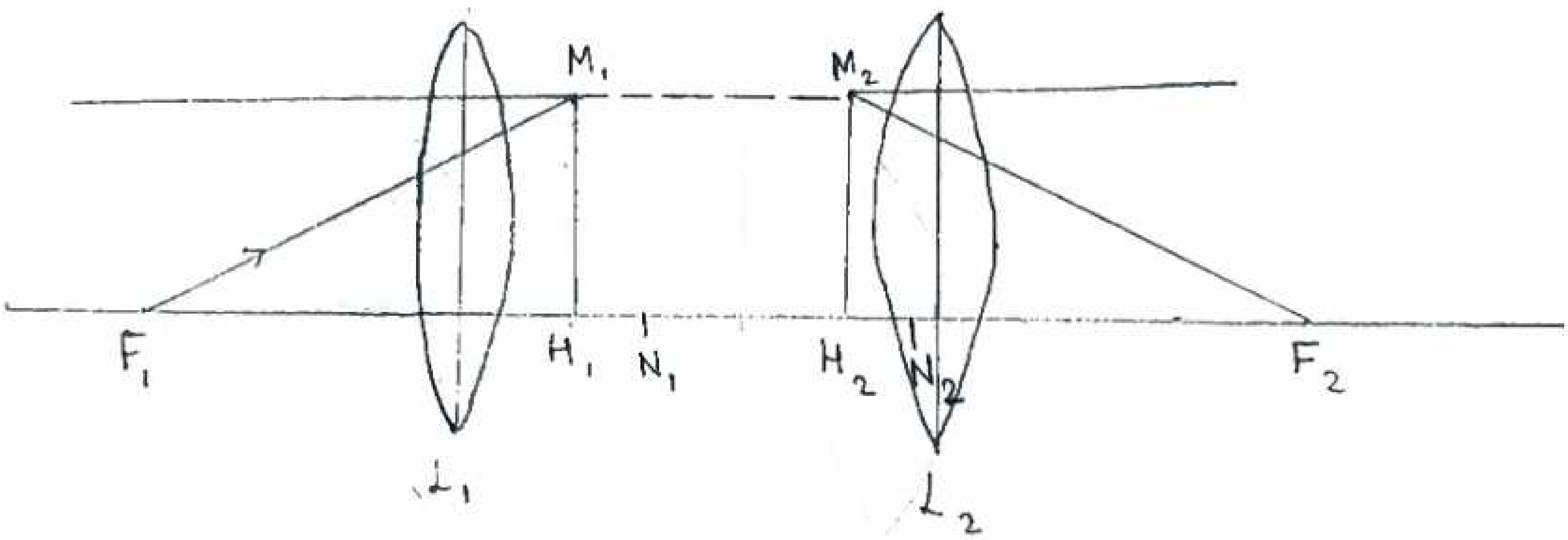
b) Two Principal Points

These are the points in the principle axis where the lateral magnification is +1, Hence any light ray striking the first principal plane at a certain height will emerge from the 2nd principal plane at same height

c) Two Nodal Point

These are two points on the principal axis that have unit positive angular magnification.

If an incident ray is directed towards the first nodal point N_1, the emergent ray will appear to come from the second nodal point (N_2) and will be exactly parallel to incident ray's original direction.



For Co-axial lens system as in diagram above -

The Principal point H_1 = Under the Point where emergent ray is parallel to principle axis is traced backwards and it meets the incident ray original path.

similarly

Principle point H_2 = Under the Point where the incident ray parallel to principal axis meets the emergent ray when traced backwards.

The Principal points F_1 and F_2 lies at a distance f = focal length of equivalent lens to left and right of Principal points H_1 and H_2 respectively.

The Nodal points are where the angular magnification is 1.

The distance between nodal points and Principal points is same.

Construction of Image Using Cardinal Points

1) The ray 1 will be taken from object parallel to principal axis. It will hit H_1 - 1st principal points' plane and emerge out the same height but will deflect after emerging towards the F_2 of equivalent lens.

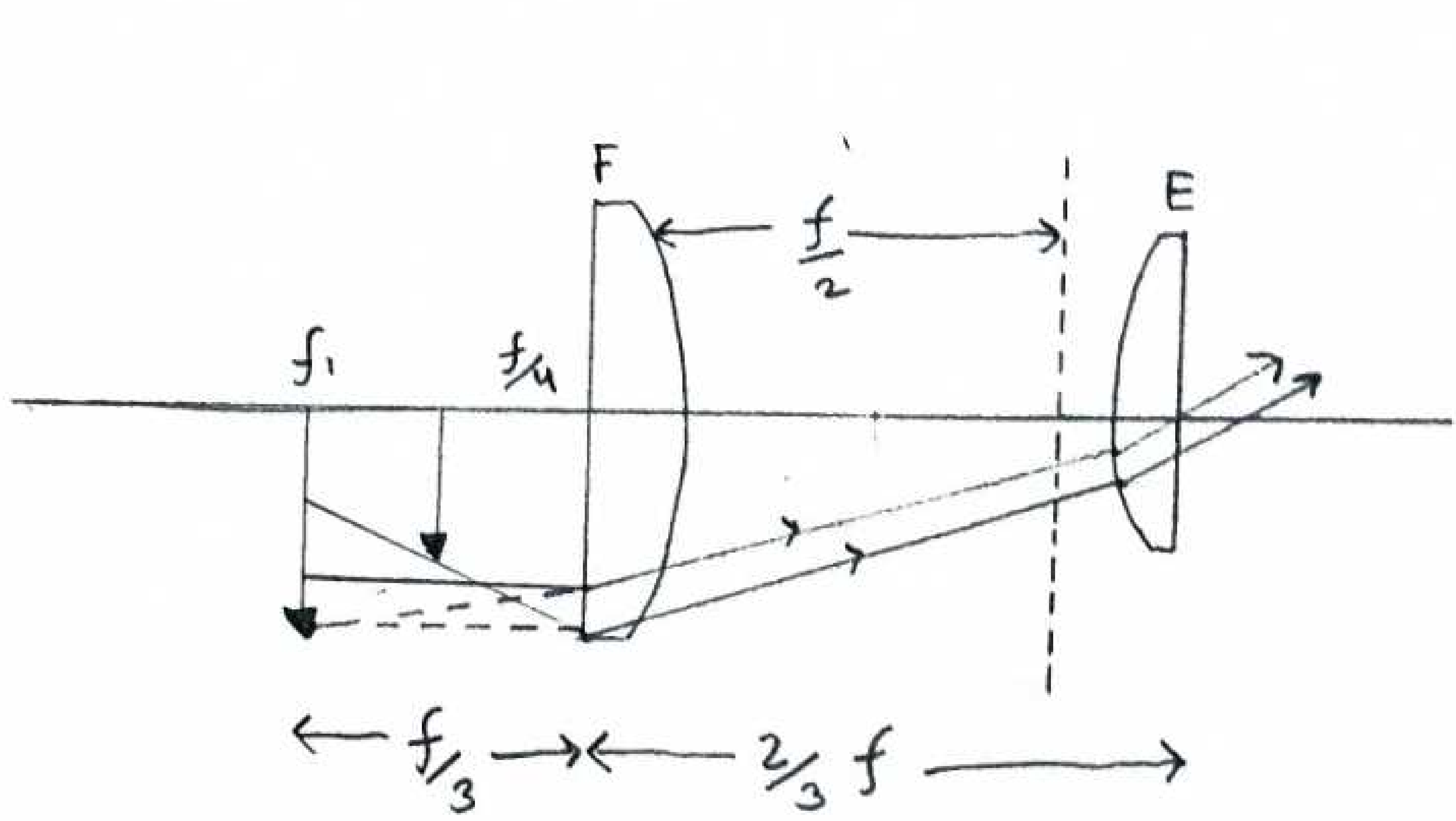
2) a second ray is directed towards the 1st nodal point and from N_2, It will emerge out with same angle.

3) A Third ray is directed from F_1, after refraction it becomes parallel to principal axis

A point of intersection of any of the above two refracted rays will give us the image of the Object thru Co-axial lens system.

Q4 Describe the Construction and Working of a Ramsden eyepiece. Find the Position and Mention the Utility of Cross Wires.

Ans It Consists of two plano Convex lenses each of focal length f, seperated by a distance equal to $\frac{2}{3}f$. The lenses are placed with their Curved surfaces facing each other as shown in image and the eyepiece is placed beyond the image formed by the Objective. This eye piece is provided with Cross wires and is used in all optical instruments where accurate quantitative measurments are required.



Ray diagram - The objective forms an image of a distance object at f. The eyepiece further forms its final image of I.. The course of rays through the eye piece is shown in the ray diagram.

Equivalent focal length - If F denotes the focal length of the equivalent lens, then -

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

$$\frac{1}{F} = \frac{1}{f} + \frac{1}{f} - \frac{2/3 f}{f^2} = \frac{4}{3f}$$

$$F = \frac{3}{4} f$$

Position of equivalent lens - The equivalent lens of focal length $\frac{3}{4} f$ must be placed behind the field lens at a distance

$$\alpha = \frac{F \times d}{f} = \frac{\frac{3}{4} f \times \frac{2}{3} f}{f} = \frac{f}{2}$$

Thus, the equivalent lens lies between the field and the eye lens.

Position of Cross-Wire.

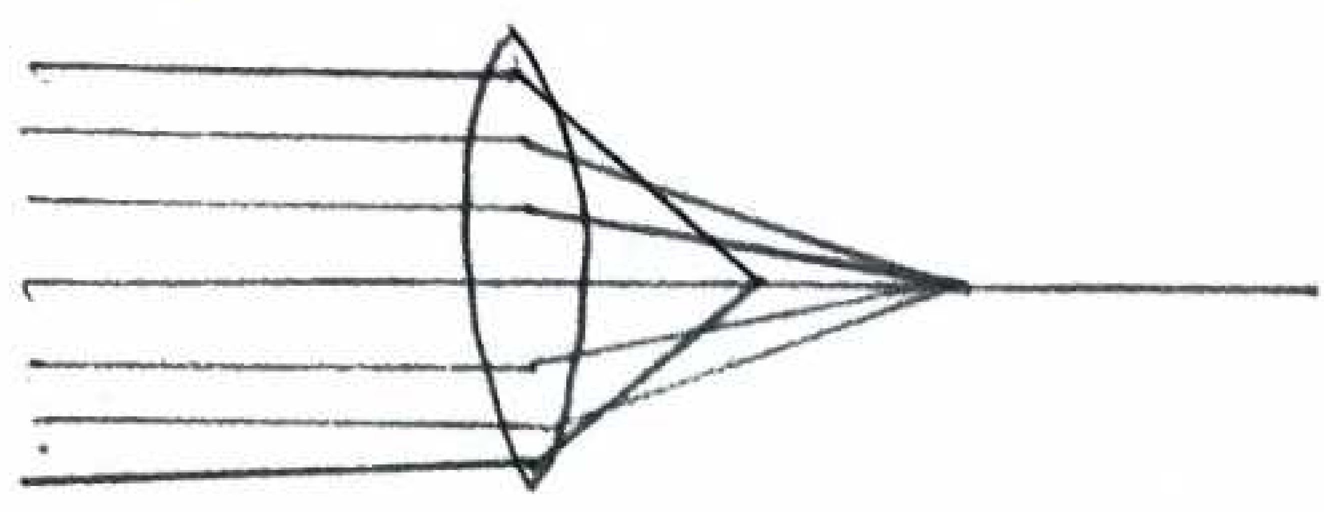
As the focal length of the eye-piece is $\frac{3}{4}f$. The image of the object due to the objective must be formed at a distance $\frac{3}{4}f - \frac{1}{2}f = \frac{1}{4}f$ in front of the field lens at I. This image will act as an object for the eye piece and the final image will be formed at infinity. Thus the cross wires must be placed at the position of the Image I, ie at a distance of $\frac{1}{4}f$ in front of the field lens F. This is the advantage of Ramsden's eye piece.

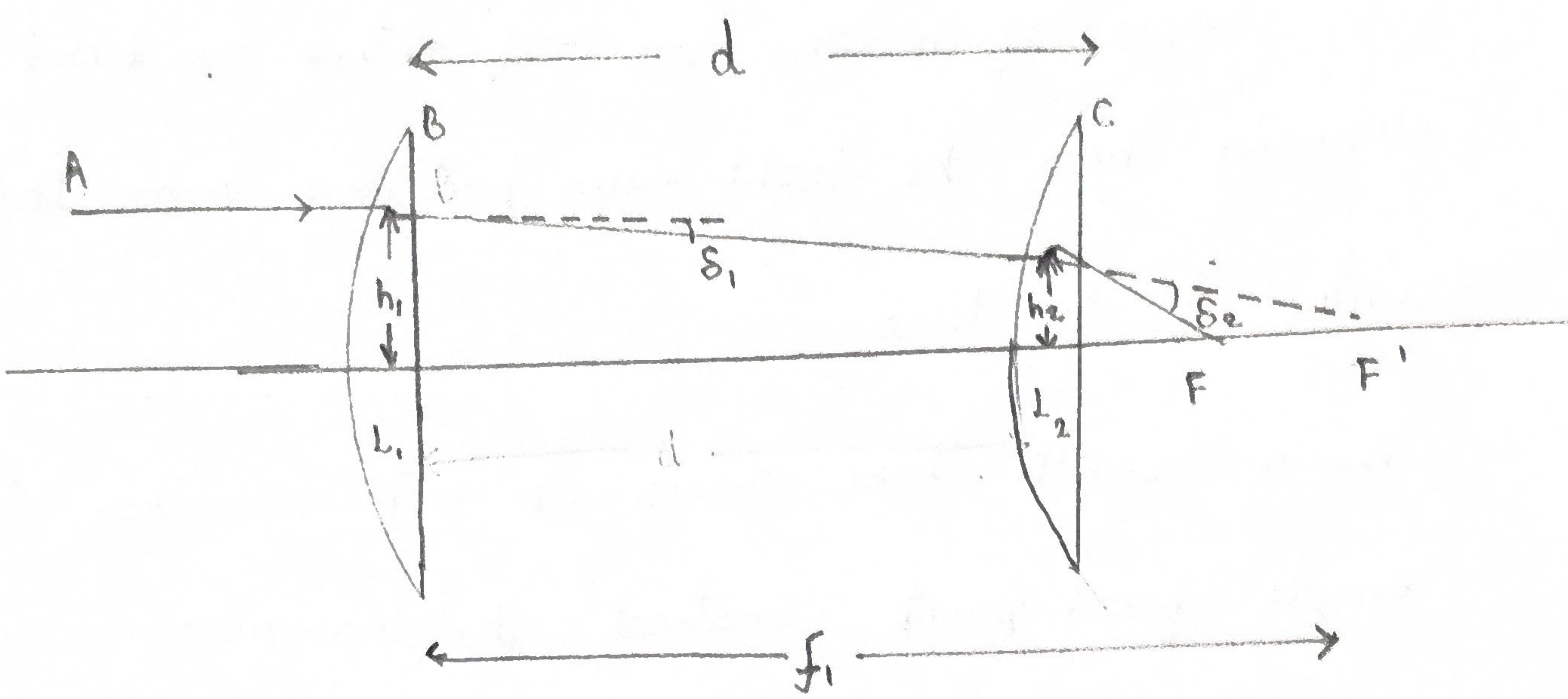
Since the chromatic abberration is small as both the lenses are made of a combination of a crown and flint glass to eliminate chromatic abberation. As both the lenses are plano convex with their convex. surfaces facing each other. The spherical abberration produced is also small.

Q5 What is Spherical aberration? Deduce the Condition for two thin lenses seperated by a finite distance

Ans Spherical aberration is an optical defect observed in lenses and mirror with spherical surfaces. It occurs because light rays passing through the outer edges of a lens are refracted or bent more strongly than the light rays passing near the centre of the lens.

As a result these rays do not converge at a single focal point. Instead, the marginal rays focus closer to the lens, while the paraxial rays focus further away. This inability to focus all incoming parallel light to a single point cause a blurred or distorted image.





A ray parallel to axis meets L_1 at height h_1 and suffers deviation $\delta_1 = \frac{h_1}{f_1}$ and is directed towards F_1', the second focal point of L_1. The reflected ray BC meets the lens L_2 at height h_2 and suffers deviation $\delta_2 = h_2 / f_2$

For minimum spherical deviation $\delta_1 = \delta_2$.

Condition for Minimum Spherical aberration for two thin lenses

Assume L_1 and L_2 are two thin lenses separated by a distance d.

f_1, f_2 are the focal length of L_1 and L_2 respectively

A ray of light ‖ to principal axis strikes the first lens L_1 at height h_1 from the principal axis.

$$\delta_1 = \frac{h_1}{f_1} \qquad\qquad \delta_2 = \frac{h_2}{f_2}$$

$$\frac{h_1}{f_1} = \frac{h_2}{f_1 - d} \qquad\qquad \frac{h_1}{f_1} = \frac{h_2}{f_2}$$

$$\Rightarrow \quad h_2 = h_1\left(\frac{f_1 - d}{f_1}\right) \qquad\qquad \frac{h_1}{f_1} = h_1\frac{(f_1 - d)}{f_1 f_2}$$

$$f_2 = f_1 - d$$

$$d = f_1 - f_2$$

Therefore, to minimize spherical aberration for a parallel incident beam, the distance separating for 2 thin lenses should be equal to difference between their focal lengths.